

EDUCATION

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- **Nanjing University** Nanjing, China
Ph.D. - Meteorology Sep 2020 - June 2026
Advisor: Prof. Lili Lei
 - **Nanjing University** Nanjing, China
B.S. - Atmospheric Science Sep 2016 - Jun 2020

RESEARCH INTERESTS

My research focuses on paleoclimate data assimilation (PDA), climate field reconstruction, and the integration of artificial intelligence (AI) with data assimilation. I am currently working on climate reconstructions of the Last Millennium to improve our understanding of climate variability and dynamics across multiple timescales.

1. Offline PDA: Paleoclimate data assimilation is most commonly implemented in an *offline* framework, where geological proxy records are combined with pre-computed climate model simulations. I developed an Integrated Hybrid Ensemble Kalman Filter (IHEnkF) that combines the strengths of small-scale flow-dependent error statistics and large-scale climatological information, providing a computationally efficient framework while maintaining high reconstruction skill.

2. Online PDA: By replacing the numerical climate model with an AI-based surrogate model, I developed an *online* PDA framework based on IHEnkF that propagates proxy information forward in time. Furthermore, I proposed an Integrated Hybrid Ensemble Kalman Smoother (IHEnkS), which assimilates future proxies to further improve the reconstruction of past climate states.

SELECTED PUBLICATIONS

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- **Sun, H.**, L. Lei, Z. Tan, L. Ning, and Z. Liu, 2026 "An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.* (Minor Revision).
 - Lei L., **Sun, H.**, Z. Tan, L. Ning, and Z. Liu, 2026 "Strongly Coupled Data Assimilation for Paleoclimate Reconstruction Using Deep Learning-Based Models.", *J. Adv. Model. Earth Syst.*, **18**, e2025MS005686, <https://doi.org/10.1029/2025MS005686>.
 - **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2025 "An Online Paleoclimate Data Assimilation with a Deep Learning-based Network", *J. Adv. Model. Earth Syst.*, **17**, e2024MS004675, <https://doi.org/10.1029/2024MS004675>.
 - **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2024 "A Hybrid Gain Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **16**, e2022MS003414, <https://doi.org/10.1029/2022MS003414>.
 - **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2022 "An Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **14**, e2021MS002674, <https://doi.org/10.1029/2021MS002674>.
 - Z. Wang, **Sun, H(co-first-author)**, L. Lei, Z. Tan, and Y. Zhang, 2023 "The Importance of Data Assimilation Components for Initial Conditions and Subsequent Error Growth", *Sci China Earth Sci.*, **67**, 105-116, <https://doi.org/10.1007/s11430-023-1229-7>.

PROJECT MANAGEMENT

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- National Natural Science Foundation of China(Basic Research Program for Young Students), 2024-2026 "Ensemble-based Multi-scale Data Assimilation for Last Millennium Climate Reconstruction", **leader**
 - National Key Research and Development Program of China, 2023-2028 "Data Assimilation Theory and Techniques for Holocene Climate Change in the Tropical Western Pacific" participation

TEACHING

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- **Teaching Assistant** Nanjing University
Graduate course: Data Assimilation Spring 2022
 - **Teaching Assistant** Nanjing University
Undergraduate course: Dynamic Meteorology Fall 2021

AWARDS

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- National Scholarship - 2024, CNY 30000
 - Presidential Scholarship - 2020, CNY 20000
 - Environmental Planning Scholarship - 2022, CNY 5000

PROGRAMMING

MATLAB/Python(numpy, scipy, xarray, Tensorflow, Pytorch)/Fortran/Linux/C